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Semester 1

Theory, Mathematics, Algorithms, and Examination Preparation

IMPORTANT — Curriculum baseline

This is a standard introductory Machine Learning curriculum. The official university syllabus, examination date, and current knowledge level were not provided, so the guide is adjustable and does not invent university-specific requirements or exam predictions. This edition intentionally focuses on Machine Learning theory and exam preparation rather than programming.

Student profile

Profile	Study priority
Semester 1 learner	First-principles understanding
3 hours available daily	Theory, mathematics, algorithms, interpretation

Contents overview

Page	Focus
1	Cover, study method, recall, and revision
2	Goals, prerequisites, diagnostic, and checklist
3	Roadmap, introduction, and mathematics
4	Supervised learning algorithms
5	Unsupervised learning and model selection
6	Evaluation and formula sheet
7	Exam practice and numericals
8	Timetable, tracking, glossary, and final review

How to use this guide

1. Preview the objective and explain the concept aloud in simple words.
2. Study one worked example, then solve a similar problem without looking.
3. Use active recall: close the guide and reproduce definitions, formulas, and algorithm steps.
4. Use spaced repetition: review after one day, three days, one week, and two weeks.
5. Finish with an exam-study workflow: learn → recall → practise → interpret → correct → revisit.

QUICK REVISION

Every strong answer connects the problem, data, method, assumptions, evaluation, and interpretation.

2. Goals, prerequisites, diagnostic, and syllabus checklist:

Learning goals — CORE

- ☐ Explain core Machine Learning concepts, vocabulary, workflows, and learning types.
- ☐ Solve numerical problems involving statistics, probability, matrices, distances, and evaluation metrics.
- ☐ Explain algorithm intuition, assumptions, steps, advantages, limitations, and use cases.
- ☐ Interpret model results, errors, metrics, clusters, principal components, and limitations.
- ☐ Write structured examination answers with definitions, diagrams, formulas, examples, and conclusions.

Prerequisite checklist

Area	Expected foundation	Check
Algebra	Rearrange equations, powers, roots, slopes, and summation notation	☐
Graphs	Read axes, trends, slopes, scatter, outliers, and distributions	☐
Statistics	Mean, median, variance, standard deviation, and samples	☐
Probability	Events, conditional probability, independence, and Bayes intuition	☐
Reasoning	Compare evidence, identify assumptions, and explain cause versus association	☐

Short diagnostic — attempt before study

1. Find the mean and median of 2, 4, 4, 8, 12.
2. If $P(A) = 0.5$ and $P(B|A) = 0.4$, find $P(A \text{ and } B)$.
3. Describe the difference between a feature, label, observation, and model parameter.
4. Explain why a test set should remain unseen until final evaluation.
5. Name one reason a model can perform well on training data but poorly on new data.

Compact syllabus checklist

Module	Status	Evidence of learning
1. Introduction and workflow	Not Started	Explain the stages and terminology
2. Mathematical foundations	Not Started	Solve formulas and interpret results
3. Supervised learning	Not Started	Compare algorithms and trace steps
4. Unsupervised learning	Not Started	Explain clusters and reduced dimensions
5. Evaluation and selection	Not Started	Choose metrics and justify a model
6. Examination preparation	Not Started	Complete timed practice and corrections

IMPORTANT

Use only these status labels: Not Started, Learning, Needs Revision, and Completed. Mark Completed only after recall and evidence.

3. Roadmap, introduction, and mathematical foundations:

Logical sequence: problem → data → features and labels → learning type → algorithm → evaluation → interpretation → communication.

Unit roadmap ⚙️

Stage	Focus	Checkpoint
1	Purpose, vocabulary, data, and workflow	Explain the complete process
2	Statistics, probability, vectors, matrices, and gradients	Solve a short numerical set
3	Regression and classification	Compare supervised methods
4	Clustering, PCA, and association rules	Interpret discovered structure
5	Validation, metrics, tuning, and selection	Defend an evaluation choice

Module 1 — Introduction to Machine Learning 🧠

AI versus ML versus Deep Learning: AI is the broad field of intelligent behaviour; ML learns patterns from data; Deep Learning uses layered neural networks to learn representations. Each narrower area is part of the broader idea.

Learning types: supervised learns from labelled examples; **unsupervised** finds structure without target labels; **reinforcement** learns actions through rewards and penalties.

Core vocabulary: a **feature** is an input variable, a **label** is the target answer, an **observation** is one example, a **parameter** is learned, and a **hyperparameter** is selected before or around training.

Workflow: define the objective → understand the data → prepare representations → divide data → learn patterns → validate → test → interpret and communicate.

Module 2 — Mathematical foundations

Concept	Meaning in ML
Mean and median	Mean gives the arithmetic centre; median gives the middle ordered value and is less affected by extreme values.
Variance and standard deviation	Variance measures average squared spread; standard deviation is its square root in the original units.
Probability	Measures uncertainty; conditional probability updates belief when evidence is known.
Vectors and matrices	Vectors represent observations or parameters; matrices arrange many vectors and support transformations.
Derivatives and gradients	A derivative measures change in one direction; a gradient collects partial derivatives and points toward steepest increase.

Essential meanings: in $\mu = \Sigma x_i/n$, μ is the mean, x_i is one value, n is the number of values, and Σ means add all values. In $\sigma^2 = \Sigma(x_i - \mu)^2/n$, σ^2 is population variance and σ is standard deviation.

Worked example: for 2, 4, 6: $\mu = (2 + 4 + 6)/3 = 4$; $\sigma^2 = [(-2)^2 + 0^2 + 2^2]/3 = 8/3$; $\sigma = \sqrt{(8/3)} \approx 1.63$. The values typically lie about 1.63 units from the mean.

EXAM TIP

Define every symbol before substitution, show each arithmetic step, and finish by explaining what the result means.

4. Module 3 — Supervised learning 🤖:

Supervised learning uses examples with known labels. **Regression** predicts a number; **classification** predicts a category or probability.

Algorithm comparison

Method	Definition and intuition	Core working steps	Advantage	Limitation / use
Linear regression	Fits a line or hyperplane to predict a continuous value.	Choose coefficients → calculate predictions → reduce squared errors.	Simple and interpretable.	Sensitive to outliers and nonlinearity; price estimation.
Multiple linear regression	Extends linear regression to several features.	Combine weighted features → estimate intercept and coefficients → interpret effects carefully.	Shows several measured relationships.	Correlated features complicate interpretation; demand prediction.
Logistic regression	Maps a weighted score to a probability for classification.	Compute score → apply sigmoid → choose threshold → assign class.	Clear baseline and probabilities.	Linear boundary; binary risk screening.
Decision tree	Uses a sequence of if–then splits to partition examples.	Choose a split → divide data → repeat → stop or prune.	Readable nonlinear rules.	Can overfit and change with small data; approval decisions.
k-nearest neighbors	Predicts from the labels of nearby observations.	Choose k → measure distance → select neighbors → vote or average.	Easy local intuition.	Needs suitable scale and can be slow; pattern recognition.
Naive Bayes	Uses Bayes' rule with a simplifying conditional-independence assumption.	Estimate class priors and likelihoods → combine evidence → choose most probable class.	Fast and effective with limited data.	Independence assumption may be unrealistic; document classification.
Support vector machine	Finds a boundary with the widest margin between classes.	Represent points → identify support vectors → maximise margin → classify by side of boundary.	Strong in high-dimensional spaces.	Scaling and parameter choices matter; image or text categories.
Random forest	Combines many varied decision	Sample data and features → grow	Stable nonlinear performance.	Less transparent

	trees and aggregates their	trees → vote or average		than one tree; risk or tabular

Exam-focused points

- **Always state the target:** continuous for regression, categorical for classification.
- **Explain assumptions:** linearity, independence, distance meaning, probability assumptions, scaling, or interpretability.
- **Trace the method:** initial input → central operation → learned output → prediction or interpretation.
- **Compare fairly:** include accuracy, interpretability, data needs, computational cost, and failure modes.

CORE

A strong algorithm answer contains definition, intuition, steps, one advantage, one limitation, and one appropriate use case.

5. Modules 4–5 — Unsupervised learning and model selection 🔍:

Module 4 — Unsupervised learning

Clustering groups similar observations without labels. **Feature extraction** transforms raw measurements into more useful representations while trying to preserve important information.

k-means steps: choose k centroids → assign each point to the nearest centroid → recompute each centroid as the cluster mean → repeat until assignments stabilise. It favours compact, roughly spherical groups and is sensitive to scale and the selected k.

Hierarchical clustering: start with individual points → repeatedly merge the closest groups → display the nested history as a dendrogram → select a level to obtain clusters. The distance and linkage rule affect the result.

DBSCAN — OPTIONAL: choose a neighbourhood radius and minimum density → identify dense core points → expand connected regions → mark isolated points as noise. It can find irregular shapes but is sensitive to density settings.

PCA: centre or standardise features → find directions of greatest variance → rank principal components → project observations into fewer dimensions. It supports compression and visualisation, but components may be difficult to interpret.

Association rule learning: discovers items that occur together. Association is not proof of causation; support measures frequency, confidence measures conditional reliability, and lift compares the rule with chance expectation.

Module 5 — Validation and model selection

Data roles: training data estimates parameters; validation data compares settings and models; the test set is held back for an honest final estimate.

Overfitting memorises noise and performs poorly on new data. **Underfitting** is too simple to capture useful structure. The **bias–variance trade-off** balances systematic simplicity error against sensitivity to sampling variation.

Cross-validation: divide training data into folds → train on most folds → validate on the remaining fold → rotate the validation fold → average results. Use it to compare settings without using the final test set.

Hyperparameter tuning: define plausible settings → compare them with validation or cross-validation → select using the stated metric → retrain on permitted data → evaluate once on the test set.

Method	Output	Strength	Watch for
k-means	Cluster assignments and centroids	Fast compact grouping	k, scale, outliers, shape
Hierarchical	Dendrogram and groups	Nested structure	Distance and linkage
DBSCAN — OPTIONAL	Dense groups and noise	Irregular shapes	Density settings
PCA	Principal components	Compression and visualisation	Scaling and meaning of components
Cross-validation	Average validation estimate	Uses limited data efficiently	Still must protect the test set

OPTIONAL

DBSCAN and deeper association-rule measures are enrichment topics unless the official syllabus requires them.

6. Model evaluation and formula sheet 📊:

Confusion matrix

Actual / predicted	Positive	Negative	Meaning
Positive	TP	FN	Correct positive / missed positive
Negative	FP	TN	False alarm / correct negative

Metric	Formula and symbols	Choose when
Accuracy	(TP + TN)/(TP + TN + FP + FN); total correct divided by all cases.	Classes and error costs are reasonably balanced.
Precision	TP/(TP + FP); among predicted positives, proportion truly positive.	False positives are costly.
Recall	TP/(TP + FN); among actual positives, proportion found.	False negatives are costly.
F1-score	2PR/(P + R), where P is precision and R is recall.	Both precision and recall matter.
ROC and AUC	ROC compares true-positive rate with false-positive rate across thresholds; AUC summarises ranking ability from 0 to 1.	Compare probability rankings across thresholds.

Regression, distance, and learning formulas

Formula	Symbols	Meaning
$MSE = \Sigma(y_i - \hat{y}_i)^2/n$	y_i actual; $\hat{y}_i$ predicted; n cases; Σ sum.	Average squared error; large errors count more.
$MAE = \Sigma\lvert y_i - \hat{y}_i\rvert/n$	$\lvert\ \rvert$ absolute value; other symbols as above.	Average absolute error in target units.
$\hat{y} = b_0 + b_1x$	$\hat{y}$ prediction; b_0 intercept; b_1 slope; x feature.	Simple linear regression prediction.
$d(a,b) = \sqrt{\Sigma(a_j - b_j)^2}$	a_j and b_j coordinates; j dimensions.	Euclidean distance between observations.
$J = \Sigma\lVert x_i - c_k\rVert^2$	x_i point; c_k assigned centroid; J objective.	k-means seeks small within-cluster squared distance.
$\sigma(z) = 1/(1 + e^{-z})$	z weighted score; e exponential constant; $\sigma(z)$ probability.	Maps a score to a value between 0 and 1.
$z = (x - \mu)/\sigma$	x value; μ mean; σ standard deviation.	Standardisation gives mean 0 and spread 1.

Essential statistics and probability

Mean: $\mu = \Sigma x_i/n$. **Sample variance:** $s^2 = \Sigma(x_i - \bar{x})^2/(n - 1)$, where $\bar{x}$ is the sample mean. **Conditional probability:** $P(A|B) = P(A \text{ and } B)/P(B)$, where A and B are events and P(B) is nonzero. **Bayes:** $P(A|B) = P(B|A)P(A)/P(B)$.

Worked calculations: for TP = 30, TN = 50, FP = 10, FN = 10: accuracy = 80/100 = 0.80; precision = 30/40 = 0.75; recall = 30/40 = 0.75; F1 = 0.75. For actual values 2, 5 and predictions 3, 4: errors are 1 and −1, so MSE = 1 and MAE = 1.

Common mistakes

- Leakage: allowing information from validation or test data to influence training or selection.
- Class imbalance: reporting high accuracy while missing the minority class.
- Association versus causation: treating co-occurrence as proof of cause.
- Uninterpreted scores: reporting a metric without a baseline, data split, threshold, or error meaning.

EXAM TIP

Choose a metric from the consequence of errors, not from habit. State what the metric rewards and what it may hide.

7. Exam practice and numericals:

Practice note: These are study prompts, not predictions of examination questions. Answer with definition → reasoning or formula → working → interpretation.

Easy — recall and recognition

- Short answer: distinguish AI, Machine Learning, and Deep Learning.
- Fill in the blank: a variable used as input to a model is a _____.
- True or false: unsupervised learning requires a labelled target for every observation.
- MCQ: which method groups unlabelled observations around centroids? A. k-means B. logistic regression C. Naive Bayes D. linear regression.
- Algorithm tracing: list the first three actions in k-means after k is chosen.
- Model interpretation: explain what a positive regression coefficient suggests, without claiming causation.

Intermediate — application and calculation

1. Find the mean, population variance, and standard deviation of 2, 4, 4, 6.
2. If P(A) = 0.6 and P(B|A) = 0.5, calculate P(A and B).
3. For A = [[1,2],[3,4]] and B = [[2,0],[1,2]], find A + B and the first row of AB.
4. For $\hat{y} = 2 + 3x$, find the prediction at x = 4 and residual when y = 15.
5. Find the Euclidean distance between (1,2) and (4,6).
6. Given TP = 30, TN = 50, FP = 10, FN = 10, calculate accuracy, precision, recall, and F1-score.
7. For actual values 2, 5 and predictions 3, 4, calculate MSE and MAE.

Advanced — explanation and synthesis

- Long answer: compare decision trees, k-nearest neighbors, and random forests by intuition, assumptions, strengths, and limitations.
- Case study: a disease-screening model misses many positive cases. Which metric should be prioritised and why?
- Model selection: explain how cross-validation and a protected test set support an honest comparison.
- Algorithm tracing: two centroids are given; assign four points, recompute centroids, and state when the process stops.
- PCA intuition: explain why the first component may preserve high variance but still be difficult to name.
- True or false: a high training score proves that a model will generalise. Correct the statement.

Compact answer key and guidance

Prompt	Answer	Reasoning cue
Mean of 2,4,4,6	4	Add values and divide by 4.
Variance of 2,4,4,6	2	Squared deviations are 4,0,0,4; divide by 4.
P(A and B)	0.3	Multiply P(A) by P(B|A).
Prediction and residual	$\hat{y} = 14$; residual = 1	Residual = actual − prediction.
Distance (1,2) to (4,6)	5	$\sqrt{(3^2 + 4^2)}$.
Confusion-matrix set	Accuracy 0.80; precision 0.75; recall 0.75; F1 0.75	Write the denominator for each metric.
MSE and MAE	1 and 1	Errors are 1 and −1; square or take absolute value.

QUICK REVISION

Redo every missed question after one day without looking at the answer, then write the exact reason the first attempt failed.

8. Timetable, self-assessment, tracker, glossary, and final checklist:

Adjustable eight-week plan — 3 hours per day

Week	Focus	Weekly evidence
1	Theory 60m; mathematics 40m; recall 30m; revision 30m; guided questions 20m	Vocabulary, workflow, diagnostic corrections
2	Statistics and probability 70m; algorithm tracing 40m; recall 30m; review 40m	Formula sheet and worked examples
3	Regression 60m; numericals 50m; recall 30m; comparison answers 40m	Regression explanation and calculations
4	Classification 60m; algorithm tracing 45m; recall 30m; revision 45m	Supervised-method comparison
5	Clustering and PCA 60m; mathematics 45m; recall 30m; review 45m	Unsupervised-method summary
6	Evaluation 60m; metrics 50m; recall 30m; model selection 40m	Metric-choice explanations
7	Mixed theory 45m; numericals 45m; case studies 45m; mock test 45m	Timed mock and corrections
8	Weak-topic repair 45m; formula recall 45m; mock test 60m; final review 30m	Final checklist and confidence update

Daily rhythm: theory 60 minutes • mathematics/numericals 40 minutes • algorithm tracing 30 minutes • active recall 25 minutes • revision 25 minutes. Use one review day and one catch-up day each week; replace the mock block with catch-up when a topic remains below confidence 3/5.

Self-assessment bands

Result	Next action
80–100%	Maintain with spaced review and mixed challenge questions.
60–79%	Review errors and retest within one week.
40–59%	Relearn definitions, formulas, and guided examples.
Below 40%	Return to prerequisites and study one small concept at a time.

Blank progress tracker

Module / Topic	Status	Confidence	Quiz performance	Revision need	Next activity	Date
_____	Not Started / Learning / Needs Revision / Completed	__/5	__%	_____	_____	_____
_____	Not Started / Learning / Needs Revision / Completed	__/5	__%	_____	_____	_____
_____	Not Started / Learning / Needs Revision / Completed	__/5	__%	_____	_____	_____

Compact glossary

Term	Meaning
Feature	Input variable used by a model.
Label	Known target attached to an example.
Parameter	Value learned from data.
Hyperparameter	Setting selected outside the learned parameters.
Loss	Numerical measure of prediction error.
Generalisation	Performance on new, unseen data.
Leakage	Improper information crossing into training or selection.
Inference	Using a trained model to produce an output.

Final examination checklist — CORE

- [ ] Define AI, ML, Deep Learning, features, labels, parameters, hyperparameters, and inference.
- [ ] Explain supervised, unsupervised, and reinforcement learning with examples.
- [ ] Solve statistics, probability, matrix, distance, regression, clustering, and metric formulas.
- [ ] Compare algorithms using intuition, steps, assumptions, advantages, limitations, and use cases.
- [ ] Explain data splits, leakage, overfitting, underfitting, bias, variance, and cross-validation.
- [ ] Interpret model results and state what a score does and does not prove.
- [ ] Complete a timed mock, correct every error, and repeat weak topics using spaced repetition.

Future study modes

- **Teach a topic:** explain from first principles with a small example.
- **Doubt clearing:** resolve one concept, formula, or interpretation step at a time.
- **Quiz me:** ask questions and reveal answers after an attempt.
- **Exam preparation:** create structured answers, timed practice, and revision priorities.
- **Quick revision:** summarise definitions, formulas, and algorithm steps.
- **Assignment/project support:** clarify theory, scope, assumptions, and interpretation.
- **Study planning:** adjust the timetable to available time and weak topics.
- **Progress update:** record status, confidence, quiz performance, and next activity.

QUICK REVISION

Learn the idea, show the mathematics, trace the algorithm, evaluate responsibly, explain the result, and record the next improvement.